Sk Mahiduzzaman



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Experience

University College Cork (UCC), Ireland (Remote) **Jun 2025 – Sep 2025**
AI/ML Research Intern – DICOM Malware Detection and Mitigation

- Engineered a lightweight CNN using TensorFlow for synthetic LSB-based malware detection in DICOM medical images.
- Developed a four-channel forensic feature extraction pipeline using LSB planes, horizontal/vertical transitions, and local residual analysis for pixel-level anomaly detection.
- Implemented Grad-CAM-based Explainable AI (XAI) to visualize suspicious image regions and interpret CNN predictions.

Education

Adamas University **2022 – 2026**
B.Tech in Computer Science and Engineering (AI & ML) **CGPA: 8.28**

P.P.D. High School **2022**
Higher Secondary (10+2) **92.60%**

P.P.D. High School **2020**
Secondary (10th) **73.86%**

Skills

Programming Languages: Python, C
Data Analysis & Machine Learning: Pandas, NumPy, Scikit-learn, Power BI, Feature Engineering
Deep Learning & GenAI: NLP, PyTorch, TensorFlow, Hugging Face Transformers, LangChain, RAG, LLMs, Fine-tuning
MLOps & Infrastructure: Docker, Linux (Bash), DVC, Git, GitHub, FastAPI
Databases & Cloud: MySQL, MongoDB, FAISS, AWS (EC2, S3)

Projects

Book Recommender System (Machine Learning) *GitHub Link*

- Developed a recommendation system using popularity-based and collaborative filtering to suggest books based on user preferences.
- Implemented data preprocessing and model serialization using Pandas, NumPy, Scikit-learn, and Pickle.
- Built a Flask application displaying popular and personalized book recommendations with covers, authors, and ratings.

Multimodal Indic Language Identification System (Speech + Text) *GitHub Link*

- Built a deep learning system to identify 12 Indian languages from speech and text.
- Fine-tuned & Evaluated XLS-R, Wav2Vec2, HuBERT, WavLM, mBERT, and XLM-R using audio preprocessing and data augmentation.
- Achieved 99.94% accuracy with 0.0047 loss through multimodal feature fusion, outperforming text-only (99.86%, 0.015 loss) and speech-only (98.39%, 0.1203 loss) baselines.

Automated MLOps Pipeline *GitHub Link*

- Engineered a reproducible ML pipeline using DVC for data versioning, preprocessing, model training, and artifact tracking.
- Containerized & Deployed the application using Docker and AWS (S3 and EKS), with Flask-based model serving.
- Automated workflows using Makefile and GitHub Actions, integrating Pytest and logging for reliable model lifecycle management.

Languages

English · Hindi · Bengali